



Homework 1

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Problem 1.3

We define the power set 2^A as the set containing all possible distinct subsets of A , including the empty set $\emptyset$ and A itself. For the given examples, we have:

a) For $A = \{2, 5\}$, we have

$$2^A = \{\emptyset, \{2\}, \{5\}, \{2, 5\}\}.$$

b) For $A = \{f, a, f\}$, we have

$$2^A = \{\emptyset, \{f\}, \{a\}, \{f, a\}\}.$$

This is because sets cannot contain duplicate elements. Therefore,

$$A = \{f, a, f\} = \{f, a\},$$

so $2^A = 2^{\{f, a\}}$.

c) For $A = \{\emptyset, \{z\}\}$, we have

$$2^A = \{\emptyset, \{\emptyset\}, \{\{z\}\}, \{\emptyset, \{z\}\}\}.$$

This is because A has two distinct elements, namely $\emptyset$ and $\{z\}$. Therefore, its power set contains all four possible subsets of these two elements.

To find the power set of the power set of A , 2^{2^A} , we first compute 2^A . We then treat 2^A as an ordinary set and compute its power set. For instance, in example (a), we have $A = \{2, 5\}$. If we define $B = 2^A$, then

$$B = \{\emptyset, \{2\}, \{5\}, \{2, 5\}\}.$$

Therefore,

$$2^{2^A} = 2^B,$$

which is simply the power set of B .

My code for iterating power sets

```
1 from itertools import combinations
2
3 def latex_element(element):
4     """Convert a single element to LaTeX."""
5
6     if element == EMPTY_SET:
7         return r"\varnothing"
8
9     if isinstance(element, str):
```



```
10     if element.strip() in {"\\varnothing"}:
11         return r"\\varnothing"
12
13     replacements = {
14         "\\": r"\textbackslash{}",
15         "&": r"\&",
16         "%": r"\%",
17         "$": r"\$",
18         "#": r"\#",
19         "_": r"\_",
20         "{": r"\{ ",
21         "}": r"\} ",
22     }
23
24     result = element
25     for old, new in replacements.items():
26         result = result.replace(old, new)
27
28     return r"\text{" + result + "}"
29
30     return str(element)
31
32
33 def latex_set(s):
34     """Recursively convert a Python set/frozenset into LaTeX."""
35
36     if len(s) == 0:
37         return r"\\varnothing"
38
39     elements = sorted(s, key=str)
40
41     formatted = []
42
43     for element in elements:
44
45         if isinstance(element, (set, frozenset)):
46             formatted.append(latex_set(element))
47
48         elif element == EMPTY_SET:
49             formatted.append(r"\\varnothing")
50
51         else:
52             formatted.append(latex_element(element))
53
54     return r"\{" + r",\, ".join(formatted) + r"\}"
55
56
57 # ----- Power set -----
58
```



```
59 def power_set(s):
60     """Return the power set of s as a set of frozensets."""
61
62     elements = list(s)
63
64     return {
65         frozenset(combination)
66         for r in range(len(elements) + 1)
67         for combination in combinations(elements, r)
68     }
69
70
71 # ----- Input -----
72
73 EMPTY_SET = "__EMPTY_SET__"
74
75
76 def parse_element(x):
77
78     x = x.strip()
79
80     if x in {"\\varnothing"}:
81         return EMPTY_SET
82
83     try:
84         return int(x)
85     except ValueError:
86         pass
87
88     try:
89         return float(x)
90     except ValueError:
91         pass
92
93     return x
94
95
96 def main():
97
98     print("Power Set Generator")
99     print("-----")
100
101     raw = input(
102         "Enter the elements separated by commas "
103         "(example: 2, 5, hello, world):\n> "
104     )
105
106     elements = [parse_element(x) for x in raw.split(",")]
107
108     # Remove dopubles
```



```
109 original_set = frozenset(elements)
110
111 iterations = int(
112     input(
113         "\nNumber of iterations "
114         "(1 = P(A), 2 = P(P(A)), etc.):\n> "
115     )
116 )
117
118 filename = "powerset.tex"
119
120 if not filename:
121     filename = "power_set.tex"
122
123 if not filename.endswith(".tex"):
124     filename += ".tex"
125
126 current = original_set
127
128 for i in range(iterations):
129     current = power_set(current)
130
131     print(
132         f"Iteration {i + 1}: "
133         f"{len(current)} elements"
134     )
135
136 output = latex_set(current)
137
138 with open(filename, "w", encoding="utf-8") as file:
139     file.write(output)
140
141 print(f"\nDone!")
142 print(f"Result written to: {filename}")
143
144 if len(output) < 5000:
145     print("\nLaTeX output:")
146     print(output)
147 else:
148     print(
149         f"\nThe LaTeX output is {len(output):,} characters"
150         f"long."
151     )
152
153 if __name__ == "__main__":
154     main()
```



Problem 1.4

Lemma 1.4.1 If $A \subset B$ then $A \cap B = A$.

Lemma 1.4.2 If $A \subset B$ then $A \cup B = B$.

Lemma 1.4.3 $C \setminus B = C \cap \bar{B}$

- a) $(C \setminus B) \subset (C \setminus A) \implies (C \setminus B) \cap (C \setminus A) = (C \setminus B)$
 $(C \setminus B) \cap (C \setminus A) = (C \cap \bar{B}) \cap (C \cap \bar{A}) = C \cap \bar{B} \cap C \cap \bar{A} = C \cap (\bar{B} \cap \bar{A})$
 that because union is asociative and comunative and we have the property $C \cap C = C$.

From DeMorgan property we have $\bar{B} \cap \bar{A} = \overline{B \cup A}$.

From Lemma 1.4.2 $\overline{B \cup A} = \bar{B} \implies C \cap (\bar{B} \cap \bar{A}) = C \cap \bar{B}$. And using DeMorgan property we get $C \cap \bar{B} = C \setminus B$

- b) $(A \setminus B) \setminus (B \setminus C) = A \setminus B$
 $(A \setminus B) \setminus (B \setminus C) = (A \cap \bar{B}) \cap \overline{(B \cap \bar{C})} = (A \cap \bar{B}) \cap (\bar{B} \cup \bar{\bar{C}})$
 we have $\bar{\bar{C}} = C$ because the set contain all the eleemnts that are not contained in the set that has all the element not contained in set C is C.
 $(A \cap \bar{B}) \cap (\bar{B} \cup C) = (A \cap \bar{B} \cap \bar{B}) \cup (A \cap \bar{B} \cap C) = (A \cap \bar{B}) \cup (A \cap \bar{B} \cap C)$
 By E we define the set containing evrything
 $(A \cap \bar{B}) \cup (A \cap \bar{B} \cap C) = ((A \cap \bar{B}) \cap E) \cup ((A \cap \bar{B}) \cap C) = (A \cap \bar{B}) \cap (E \cup C)$

From the property on the evrything set $\begin{cases} E \cup C = E \\ E \cap C = C \end{cases} \implies$
 $\implies (A \cap \bar{B}) \cap (E \cup C) = (A \cap \bar{B}) \cap E = A \cap \bar{B} = A \setminus B$

- c) $A \cup (A \cap B) = A$
 we allredy did this at point b) but i will do it once again here. By E we define the set containing evrything

From the property on the evrything set $\begin{cases} E \cup A = E \\ E \cap A = A \end{cases}$
 $A \cup (A \cap B) = (A \cap E) \cup (A \cap B) = A \cap (E \cup B) = A$

Problem 1.5

$$\left(\bigcap_{i=1}^n A_i \right)^c = \bigcup_{i=1}^n A_i^c$$

If X is an element of all sets A_i then $x \in \bigcap_{i=1}^n A_i \implies x \notin (\bigcap_{i=1}^n A_i)^c$ and also that means that X is in none of the sets A_i^c sets them $X \notin \bigcup_{i=1}^n A_i^c$ and both conditions are equivalent wich implies that DeMorgan indenty rule is true for X in all sets A_i .

If X is not an element of any set A_i then $x \notin \bigcap_{i=1}^n A_i \implies x \in (\bigcap_{i=1}^n A_i)^c$ and also that means that X is in at least one of the sets A_i^c them $X \in \bigcup_{i=1}^n A_i^c$ and both conditions are equivalent wich implies that DeMorgan indenty rule is true for X in none of the sets A_i .

Problem 1.6

- a) $X \times Y = \{(a, b), (a, c), (a, d), (b, b), (b, c), (b, d)\}$



- b) $Y^2 = Y \times Y = \{(b, b), (b, c), (b, d), (c, b), (c, c), (c, d), (d, b), (d, c), (d, d)\}$
 c) $Y \times X \times X = \{(b, a, a), (b, a, b), (b, b, a), (b, b, b), (c, a, a), (c, a, b), (c, b, a), (c, b, b), (d, a, a), (d, a, b), (d, b, a), (d, b, b)\}$

At point c) we just treat $Y \times X$ as a new set.

Problem 1.7

$A \times B = \{(a, b) \mid a \in A, b \in B\}$ If $A \times B = B \times A$ that means that any arbitrary ordered set (a, b) should exist in both products. So we have $(a, b) \in A \times B$ for $a \in A, b \in B$ and $(a, b) \in B \times A$ for $b \in A, a \in B$ which results that a is in A and B and b is in A and B and both a and b are any arbitrary element from A and consecutively B which means that this is possible only if $A = B$. If $A \times B = \emptyset$ either A or B should be *varnothing* because from the property of the cartesian product if $a \in \emptyset$ there are no ordered sets to pair because $\emptyset$ is an empty set. Consequently $B \times A = \{(a, b) \mid a \in B, b \in A\}$ and is again $\emptyset$ because $A = \emptyset$ so there does not exist any b element so we proved that

$$\begin{cases} A \times B = \emptyset \\ B \times A = \emptyset \end{cases} \rightarrow A \times B = B \times A = \emptyset$$



Problem 1.8

1. $f(n) = a$
 $n + 2 = a \Rightarrow n = 2 - a$ this is a straight line so it only maps to one element so is injective $f(0) = 0 + 2 = 2$, $f(n) = 0 \Rightarrow 2 + n = 0 \Rightarrow n = -2 \notin \mathbb{N}$ which means is not surjective and bijective
2. This is another straight line so is injective because it has only one solution for any $g(x) = a$. It has an interesting property in only outputs odd numbers so is not surjective and bijective.
 $g(0) = -5$, $g(n) = 0 \Rightarrow 2n - 5 = 0 \Rightarrow n = 2.5 \notin \mathbb{Z}$
3. we define $h(x)$ as $h(x) = \sum_{k=0}^{\lfloor \log_{10} x \rfloor} (\lfloor x \cdot 10^{-k} \rfloor \bmod 10)$ where $\lfloor x \rfloor$ is the lower bound of x . $h(21) = 3$ and $h(12) = 3$ so $h(x)$ is not injective. $h(x)$ can input negative numbers but due to this nature can't output negative numbers because the digits of a number are whole numbers between 0 and 9 so $h(x)$ is not surjective or bijective
4. Let's find the function minimum. $q'(n) = (n^2 - n)' = 2n - 1$ so the minimum is at point $n = 0.5$ so if we compute the adjacent point $q(0) = 0$, $q(1) = 0$ so is not injective.
 Second method $n^2 - n = n(n - 1) = 0 \Rightarrow \begin{cases} n = 0 \\ n = 1 \end{cases}$ and because we have 2 solutions it is not injective. It is neither surjective or bijective because $q(2) = 2$ and $q(3) = 6$ but $q(n) = 3 \Rightarrow n = \frac{1 \pm \sqrt{13}}{2}$ and $n \notin \mathbb{Z}$
5. $r(n_1) = r(n_2) \Rightarrow \frac{1}{n_1} = \frac{1}{n_2} \Rightarrow n_1 = n_2$ so it maps distinct elements to distinct outputs, which means is injective.
 For $q = 2 \in \mathbb{Q}^+$, $r(n) = 2 \Rightarrow \frac{1}{n} = 2 \Rightarrow n = 0.5 \notin \mathbb{N}^+$ (also $2/3 \in \mathbb{Q}^+$ has no integer pre-image) which means is not surjective and bijective.

Problem 1.9

- a) **Finite.**
 By the Fundamental Theorem of Algebra, a polynomial equation of degree 5 has at most 5 roots in $\mathbb{C}$. Therefore, the set of solutions is finite.
- b) **Countably infinite.**
 The set is

$$S = \{n^2 \mid n \in \mathbb{N}\}.$$
 The function $f(n) = n^2$ defines a bijection $f : \mathbb{N} \rightarrow S$. Therefore, S is countably infinite.
- c) **Uncountable.**
 $\mathbb{C}$ is in bijection with $\mathbb{R}^2$. Since $\mathbb{R}$ is uncountable, $\mathbb{R}^2$ is also uncountable. Therefore, $\mathbb{C}$ is uncountable.
- d) **Finite.**
 Regardless of the actual number of Romanian words, every word of length at most 10 can be formed using the finite alphabet of 31 letters. Thus, the total number of possible strings of length at most 10 is bounded by

$$\sum_{k=1}^{10} 31^k < \infty.$$



Therefore, the set of Romanian words with no more than 10 letters is finite.

e) **Depends on the sets A and B .**

- If $|A| \neq |B|$, then there are no bijections from A to B , so the set of bijections is empty and therefore finite.
- If A and B are finite with $|A| = |B| = n$, then there are exactly $n!$ bijections, so the set is finite.
- If A and B are infinite with $|A| = |B|$, then the set of bijections is uncountable.

f) **Uncountable.**

Since $B \subseteq A \cup B$ and B is uncountable, the set $A \cup B$ must also be uncountable.

g) **Uncountable.**

Lines with a fixed slope a are given by

$$y = ax + b,$$

where $b \in \mathbb{R}$. Each such line is uniquely determined by its intercept b , giving a bijection with $\mathbb{R}$. Therefore, the set of all lines with the same slope is uncountable.

h) **Countably infinite.**

Every English word can be represented as a finite string over a finite alphabet. The set of all finite strings over a finite alphabet is countably infinite. Therefore, the set of all English words is countably infinite.

i) **Uncountable.**

Every circle in $\mathbb{R}^2$ is uniquely determined by its center $(x_0, y_0) \in \mathbb{R}^2$ and its positive radius $r \in \mathbb{R}^+$. Since $\mathbb{R}^2 \times \mathbb{R}^+$ is uncountable, the set of all circles is uncountable.

j) **Finite.**

The set contains exactly

$$2^{10^{10}}$$

elements, which is a fixed finite integer. Therefore, the set is finite.

k) **Countably infinite.**

The set of all finite bit strings is

$$\{0, 1\}^* = \bigcup_{n=0}^{\infty} \{0, 1\}^n.$$

Each set $\{0, 1\}^n$ is finite, and a countable union of finite sets is countable. Since there are infinitely many finite bit strings, the set is countably infinite.

Problem 1.10

a) **Finite.**

Let

$$A = [0, 1], \quad B = [1, 2].$$

Both A and B are uncountable intervals in $\mathbb{R}$. However,

$$A \cap B = \{1\},$$

which is finite.



b) **Countably infinite.**

Let

$$A = (0, 1) \cup \mathbb{Z}, \quad B = (2, 3) \cup \mathbb{Z}.$$

Both A and B are uncountable. Their intersection is

$$A \cap B = \mathbb{Z},$$

which is countably infinite.

c) **Uncountable.**

Let

$$A = [0, 2], \quad B = [1, 3].$$

Both A and B are uncountable. Their intersection is

$$A \cap B = [1, 2],$$

which is an interval in $\mathbb{R}$ and therefore uncountable.

Problem 1.11

Assume by contradiction that $\mathbb{N}^\infty$ is countable. Then its elements can be enumerated in a list $S_1, S_2, S_3, \dots$, where each S_i is an infinite sequence of non-negative integers:

$$\begin{aligned} S_1 &= (s_{1,1}, s_{1,2}, s_{1,3}, \dots) \\ S_2 &= (s_{2,1}, s_{2,2}, s_{2,3}, \dots) \\ S_3 &= (s_{3,1}, s_{3,2}, s_{3,3}, \dots) \\ &\vdots \end{aligned}$$

We construct a sequence

$$D = (d_1, d_2, d_3, \dots)$$

and define each term by

$$d_k = s_{k,k} + 1 \quad \text{for all } k \geq 1.$$

Since $s_{k,k} \in \mathbb{N}$, it follows that $d_k \in \mathbb{N}$ for all $k \geq 1$. Thus,

$$D \in \mathbb{N}^\infty.$$

However, D differs from S_k at the k -th entry, since

$$d_k = s_{k,k} + 1 \neq s_{k,k}.$$

Therefore,

$$D \neq S_k$$

for every $k \geq 1$. Consequently, D is not in the enumeration.

This contradicts the assumption that the list contains all elements of $\mathbb{N}^\infty$. Therefore, $\mathbb{N}^\infty$ is uncountable.